LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Rocket ballistics introduction, forces acting on rocket in flight and stability
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Rocket ballistics introduction, forces acting on rocket in flight and stability
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical concepts via PPT; No hands-on practical activity; Instructor supervision required for Q&A;
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four main branches of ballistics?
(b) Name the primary forces acting on a projectile in flight.
(c) Define the term 'stability' in the context of rocket flight.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Ballistics • Internal Ballistics • External Ballistics • Terminal Ballistics • Trajectory types	L & Ds	PPT, IB	15 min
(b)	• Thrust Force • Weight Force • Drag Force • Lift Force • Equation of motion	L & Ds	PPT, PDF	20 min
(c)	• Longitudinal Stability • Lateral Stability • Spin Stabilization • Aerodynamic Center • Center of Gravity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the center of gravity affects rocket stability.
(b) Describe the role of drag force during the terminal phase of flight.
(c) Differentiate between spin stabilization and aerodynamic stabilization.

12. Home Assignment and Follow Up	Prepare a brief note on how wind conditions affect bomb ballistics as covered in LP 06.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Bomb ballistics –behaviour of bomb when released from aircraft in wind condition	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min